

Muhammad Farhan Azmine

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Education

Virginia Tech (GPA: 4.00 / 4.00)

PhD in Computer Engineering (Direct PhD; MS completed)

Blacksburg & Alexandria, VA

Aug 2022 – Dec 2026 (Expected)

Bangladesh University of Engineering and Technology (BUET)

Bachelors of Science in Electrical & Electronics Engineering

Dhaka, Bangladesh

Jan 2013 – Sep 2017

- **Relevant Coursework:** Advanced Digital Design, Advanced Computer Architecture, Testing VLSI Techniques, VLSI Device Modeling, Advanced Analog IC Design, Deep Learning, Advanced Machine Learning, Computation in Data Science

Publications

- **Muhammad F. A.,** Li, R., Sharma, G., & Yi, Y. (2025). SpikeSpec: On-Chip Learning Neuromorphic Accelerator for Spectrum Sensing. *IEEE Transaction on Computer Aided Design (TCAD)*, Feb 2025.
- Li, R., **Muhammad F. A.,** Sharma, G., & Yi, Y. (2025). Efficient Digital Architecture of Spiking Encoders. In *Proc. ISQED 2025*.
- Lin, C., **Muhammad F. A.,** Liang, Y., & Yi, Y. (2024). Neuro-Inspired AI Accelerator for 6G Networks. *Front. Comput. Neurosci.*
- Lin, C., **Muhammad F. A.,** & Yi, Y. (2023). Accelerating Wireless Communications with FPGA-Based AI. In *ICCAD 2023*.

Recent- ASIC Tapeout

- Taped out a RV32I RISC-V CPU on Sky130 PDK (21.8K cells, 334.9K μm^2 , 15.9 mW), with full ISA verif. & post-layout validation.

Research & Design Experience

Research Assistant (Functioning as RTL Engineer – Complex Digital System Design & Verification)

June 2023 – present

Supervisor : Dr. Yang (Cindy) Yi

RTL design & tapeout of RISC-V Single-Cycle CPU on Sky130 PDK & building Formal Verification Framework

[\[Github link\]](#)

Fall 2025

- Designed and verified a **single-cycle RV32I RISC-V CPU** taped out on **Sky130 PDK** (21.8K cells, 334.9K μm^2 total area, 15.9 mW total power), meeting timing across FF/SS/TT corners in **Cadence Genus**. Performed PnR, CTS & STA for GDS in **Cadence Innovus & Tempus**
- Developed a **UVM-lite environment with abstracted register/memory monitors** and **property-based formal checks** for 40+ RV32I instructions using **Cadence JasperGold** and **Xcelium**, achieving **100% proof convergence**, **98% coverage**, and **80% debug effort reduction**.
- **Reduced task flow by 80%** by creating a **Makefile script** to automate **C code** disassembly into **LLVM IR** for R32IV CPU using the **risc64-gcc-elf** toolchain in **Ubuntu Linux 24.04 OS**.

PPA optimization of Spiking Neural Network RTL architecture design with On-Chip learning for Spectrum-Sensing

[\[Github link\]](#)

Fall 22 - Spring 24

- Achieved **100% SystemVerilog Post-Si validation** through hardware–software co-integration on Zynq SoC, enabling real-time sample delivery to **RTL (PL)** and result capture via ZynQ AXI-UART (PS), with a **custom C++ parser** for CSV input processing and output generation.
- Ensured **0% state-space explosion** of updated weight register by coverage analysis & property check with **concurrent & immediate assertions**
- Reduced **RTL area by 60%** in **SystemVerilog** using resource-shared **adders** and **LUTs** via **serialization with SIPO shift registers**.
- **Throughput increase by 58 MHz** with critical path balancing between **priority encoder & exponential approximator**
- Reduced latency by 50% using simple **dual port memory ram** in read-then-write mode for weight learning update

RTL integration and Clock Domain Crossing of Ethernet-MAC IP interface with AI Accelerator inference chip

[\[Github link\]](#)

Spring 23-Fall 23

- **Boosted design frequency by 100 MHz** by implementing **Clock Domain Crossing** to achieve 200 MHz frequency for target accelerator through synchronizing with Ethernet PHY communication at 125 MHz using **Ping-pong buffer**, **CDC AXI-handshake IPs** and **Asynchronous FIFOs**
- **Improved SystemVerilog verification coverage by 30%** through modifying BIST testbench in frame data transfer between Ethernet-PHY and target accelerator by creating **over 4 protocol-variant Ethernet frame stimulus patterns** including error-injection & backpressure scenarios
- **Increased data transfer throughput 5x** by Ethernet-MAC IP integration with target accelerator design at 125 MHz frequency
- **Performed 100% ML algorithm verification** using **C++ simulator** $\langle 16, 10 \rangle$ fixed-point format with template libraries like **std::vector**
- **Reduced IP area usage in SystemVerilog RTL by 33.3%** through **DSP48E1 IP** integration at RTL-level for inference MAC operation

Technical Skills

- **Techniques:** RTL design, STA, Power optimization, Clock-Domain-Crossing, UVM, DFT, FPGA-IP Integration, ASIC implementation
- **Languages:** SystemVerilog, Verilog, Matlab (Script), Python (OOP) [\[Github link\]](#), C++, Java [\[Github link\]](#), Tcl, Linux Shell
- **Tools & Libraries:** Cadence Suite (Xcelium, Genus & JasperGold), TensorFlow [\[Github link\]](#), PyTorch, Scikit-learn [\[Github link\]](#), Vivado, Quartus
- **Concepts:** UVM, SVA Formal Verification, AXI-DMA, AXI4, UART, Ethernet-TCP/UDP, SPI, VGA, RISC V ISA, Cache memory mapping